① $U = 140\,kV$ $r_1 = 0.6\,cm$ $r_2 = 2.8\,cm$ $r_3 = 3.4\,cm$ $\varepsilon_{r_1} = 8$ $\varepsilon_{r_2} = 8$

a)

$$A = \underbrace{\frac{1}{8} \cdot \frac{2.8-0.6}{2.8 \times 0.6}}_{0.164} + \underbrace{\frac{1}{8} \cdot \frac{3.4-2.8}{3.4 \times 2.8}}_{0{,}00788} = 0{,}17188 \approx 0.17$$

$$U_1 = \underbrace{\frac{140}{0.17 \times 8}}_{102.9} \cdot \underbrace{\frac{2.8-0.6}{2.8 \times 0.6}}_{1.3} = 133.77\,kV$$

$$E_{1max} = (133.77) \cdot \frac{2.8 \times 0.6}{(2.8-0.6)} \cdot \frac{1}{(0.6)^2} = 283.75\,kV/cm$$

$$E_{1min} = (133.77) \cdot \frac{2.8 \times 0.6}{\underbrace{(2.8-0.6)}_{2.2}} \cdot \frac{1}{(2.8)^2} = 13{,}03\,kV/cm$$

$$U_2 = \underbrace{\frac{140}{0.17 \times 8}}_{102.9} \cdot \underbrace{\frac{3.4-2.8}{3.4 \times 2.8}}_{0.49} = 50{,}42\,kV$$

$$E_{2max} = (50{,}42) \cdot \frac{3.4 \times 2.8}{3.4-2.8} \cdot \frac{1}{(2.8)^2} = 102{,}04\,kV/cm$$

$$E_{2min} = (50{,}42) \cdot \frac{3.4 \times 2.8}{\underbrace{3.4-2.8}_{0.6}} \cdot \frac{1}{(3.4)^2} = 69{,}2\,kV/cm$$

b)

$E_{1max} = 133.77 > E_{b_1} = 40\,kV/cm$ layer 1 cannot withstands

$E_{2max} = 50.42 < E_{b_2} = 105\,kV/cm$ layer 2 withstands

$$E_{2\,max\,new} = 140 \cdot \frac{2.8 \times 3.4}{3.4-2.8} \cdot \frac{1}{(3.4)^2} =$$

Nurullah
Said
Toğan
150720823

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Nurullah Said Tüğmen
150720823

$K=8$
$L=8$
$M=2$
$N=3$

③ $U=60kV$ $\quad \varepsilon_r=8$
$a=8mm$
$S=15cm^2$

b) capacity: $C=\frac{\varepsilon S}{a}=\frac{\varepsilon_0\varepsilon_r S}{a}=\frac{8.854\cdot10^{-12}\times\cancel{8}\times15\times10^{-4}}{\cancel{8}\times10^{-3}}=132,81\times10^{-13}=13,281\,pF$

c) charge: $Q=CU=13.281\times10^{-12}\times60\times10^{3}=796,86\times10^{-9}=796.86\,nC$

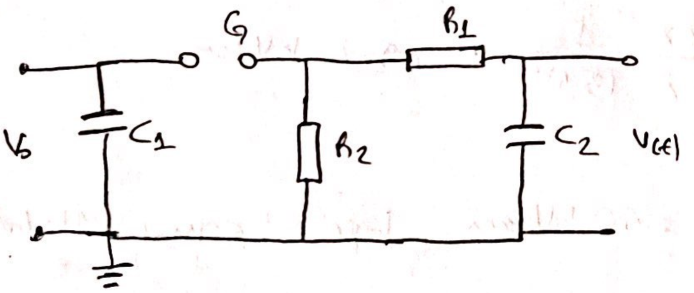
a) Electric field strength: $E=\frac{U}{a}=\frac{60\times10^{3}}{8\times10^{-3}}=7.5\times10^{6}\frac{V}{m}$

d) flux density $D=\varepsilon E=\varepsilon_0\varepsilon_r E=8.854\times10^{-12}\times8\times7.5\times10^{6}=531,24\times10^{-6}\,C/m^2$

e) max voltage:
breakdown voltage: $U_d=E_d\cdot a=45\frac{kV}{cm}\times8\times10^{-1}cm=36\,kV$ $\qquad U<U_d=36kV$
$E_d=45$

---

②



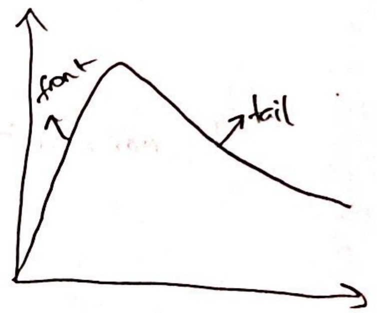
C1 will be charged from DC source and after limit, spark gap will break down.

G → like voltage-sensitive switch

R1 → for front time

R2 → for discharging the capacitor, controlling wave tail

test objects will be connected parallel to C2.



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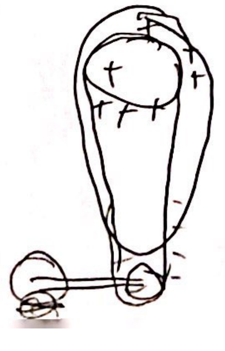
④

mechanical to electricity



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